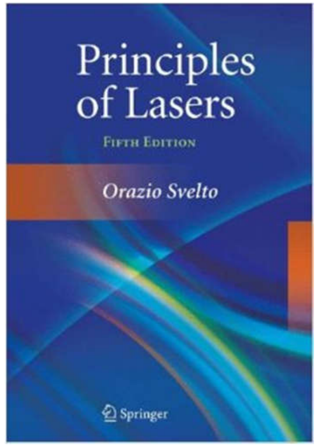
南方科技大学
SOUTHERN UNIVERSITY OF SCIENCE AND TECHNOLOGY

《激光原理与技术》

Principles and Technologies of Lasers



Ch4-3
Oscillation Characteristics of Lasers

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Chapter 4: Oscillation Characteristics

❖ 4.4 Relaxation Oscillation in Pulse Lasers

- Qualitative interpretation of relaxation oscillation
- Theoretical treatment of relaxation oscillation
- Practical significance of relaxation oscillation

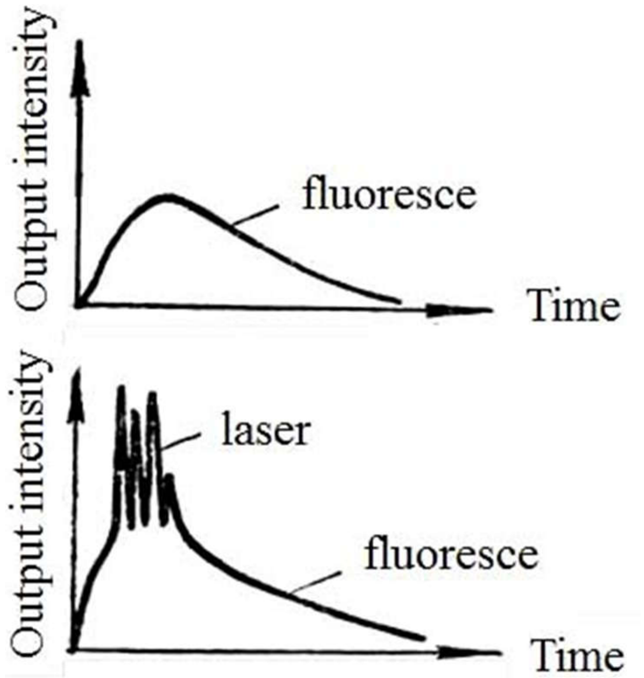
❖ 4.5 Line-Width Limit of Single Mode Lasers

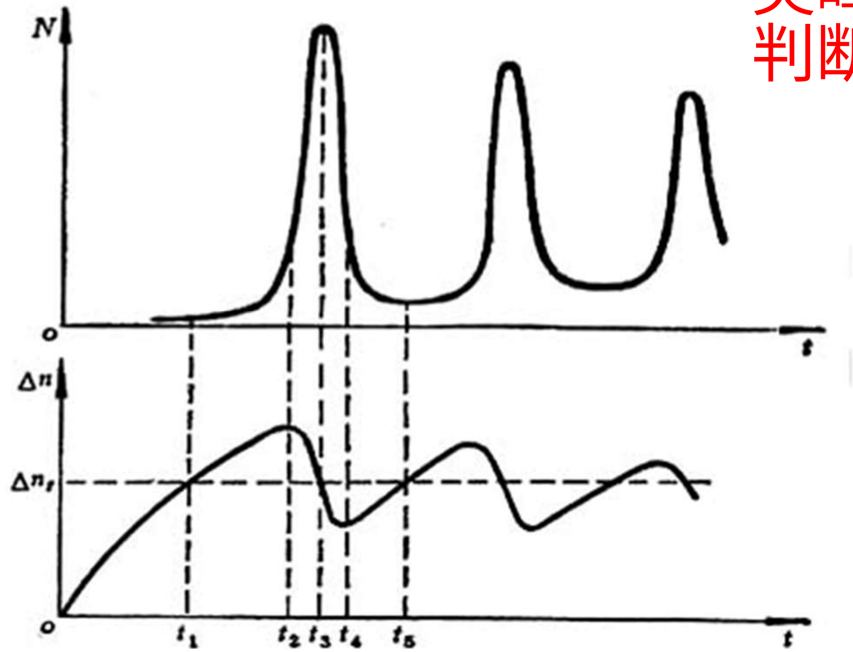
- Passive cavity line-width limit
- Active cavity line-width limit

❖ 4.6 Frequency Pulling of Laser

4.4 Relaxation Oscillation in Pulse Lasers

❖ Qualitative interpretation of the formation of relaxation oscillations (transient behaviors of Δn and N_l)





尖峰序列
判断题

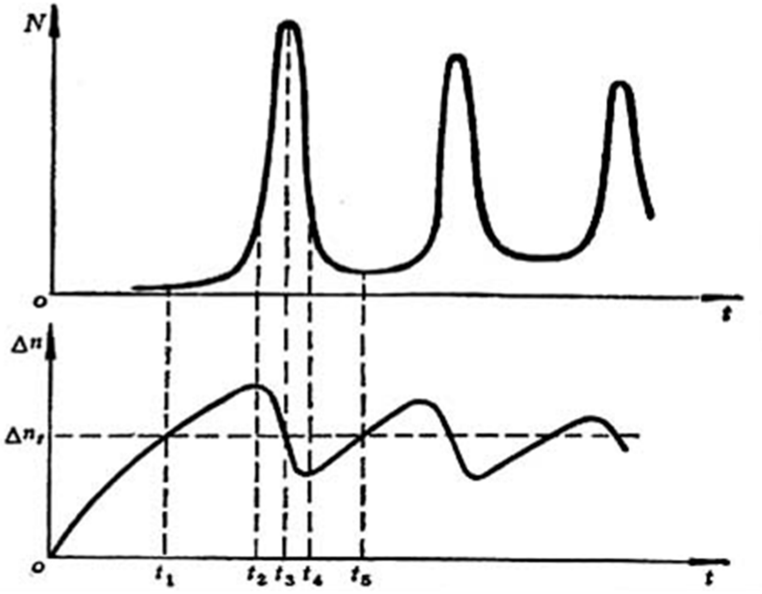
➤ t_1-t_2 The increase rate of Δn due to pumping > The decrease rate of Δn due to stimulated emission

$$\Delta n\uparrow \rightarrow \Delta n=\Delta n_t\left(t=t_1\right)\rightarrow N_l\uparrow,\Delta n\uparrow\left(t_1<t<t_2\right)$$

- t_2-t_3 Stimulated radiation causes N_l to rise sharply $\rightarrow \Delta n \downarrow$

$$\frac{dN_l}{dt} = -\frac{d\Delta n}{dt}\left(t = t_2\right) \rightarrow \begin{matrix} N_l \uparrow \\ \Delta n \downarrow \end{matrix} \left(t > t_2\right)$$

$$\rightarrow \begin{matrix} N_l \\ \Delta n_t \end{matrix} \left(t = t_3\right) \quad \text{maximum}$$



- t_3-t_4 Stimulated radiation causes $\Delta n < \Delta n_t$ $N_l \downarrow$ sharply, $\Delta n \downarrow$

- t_4-t_5 $\left\{ \begin{array}{l} \text{If pump hasn't stopped yet, weakening of stimulated radiation leads to} \\ \qquad N_l \downarrow \rightarrow \Delta n \uparrow \rightarrow \Delta n = \Delta n_t \\ \text{Rate of } \Delta n \text{ reduction by stimulated radiation = Rate of } \Delta n \text{ increase by pump} \end{array} \right.$

- $t>t_5$ Start the evolution of 2nd pulse. This process occurs repeatedly during the pumping process. As a result, the laser shows a series of spikes.

- The relaxation oscillation process occurs during the establishment of a laser. During pumping, both the inverted population and photon density undergo rapid changes due to STE. As a result, the output exhibits the characteristics of relaxation oscillations, forming multiple spikes.
- The stronger the pump, the higher the damping oscillation frequency, leading to a faster decay.

❖ Theoretical treatment of the relaxation oscillation process (solve the rate equation at non-steady-state)

1) Exact solution: numerical method

✓ 2) Approximate solution: 1st order perturbation of steady-state

Qualitative interpretation is OK.

The quantitative comparison does not align well with the experimental results. This discrepancy arises because Δn and N are not merely small fluctuations around the equilibrium state; instead, they may exhibit large fluctuations or intermittent oscillations..

Example: Four-level system, single-mode $\nu=\nu_0$, $n_2 \approx \Delta n$

Assume $\eta_F = \eta_1 = \eta_2 = 1$ $\quad L = l \quad$ $\eta_1 = \dfrac{S_{32}}{S_{32} + A_{30}}$ $\quad$ $\eta_2 = \dfrac{A_{21}}{A_{21} + S_{21}}$

$$\begin{cases} \dfrac{dN(t)}{dt} = \Delta n(t)\sigma_{21}\upsilon N(t) - \dfrac{N(t)}{\tau_R} \\ \dfrac{d\Delta n(t)}{dt} = -\Delta n(t)\sigma_{21}\upsilon N(t) - \Delta n(t) A_{21} + \left[n - \Delta n(t)\right] W_{03} \end{cases}$$

In the 1st order perturbation approximation, assume that the transient photon numbers and the inverted populations fluctuate slightly near their corresponding equilibrium values.

$$N(t) = N_0 + N'(t) \qquad N'(t) << N_0$$

$$\Delta n(t) = (\Delta n)_0 + \Delta n'(t) \qquad \Delta n'(t) << (\Delta n)_0$$

N_0 and $(\Delta n)_0$ steady-state solution

$N'(t)$ and $\Delta n'(t)$ small value

(1) Steady-state ($dX/dt=0$): N_0 and $(\Delta n)_0$

$$\frac{dN(t)}{dt}=\Delta n(t)\sigma_{21}\upsilon N(t)-\frac{N(t)}{\tau_R} \Rightarrow \Delta n(t)=\frac{1}{\tau_R\sigma_{21}\upsilon}=\frac{\delta}{\sigma_{21}l}=\Delta n_t$$

$$(\Delta n)_0=\Delta n_t$$

$$\frac{d\Delta n(t)}{dt}=-\Delta n(t)\sigma_{21}\upsilon N(t)-\Delta n(t)A_{21}+\left[n-\Delta n(t)\right]W_{03}$$

$$N_0=\frac{1}{\Delta n_t\sigma_{21}\upsilon}\left[W_{03}\left(n-\Delta n_t\right)-A_{21}\Delta n_t\right]=\tau_R\left[W_{03}\left(n-\Delta n_t\right)-A_{21}\Delta n_t\right]$$

(2) Approximate solution with 1st order perturbation

$$\frac{dN'}{dt}=\left\{\left[\Delta n_t+\boxed{\Delta n'(t)}\right]\sigma_{21}\upsilon\left[N_0+N'(t)\right]\right\}-\left[\frac{N_0+N'(t)}{\tau_R}\right]$$

$$=\sigma_{21}\upsilon\left[\cancel{\Delta n_t N_0}+\cancel{\Delta n_t N'(t)}+\Delta n'(t)N_0+\underbrace{\Delta n'(t)N'(t)}_{\approx 0}\right]-\cancel{\frac{N_0}{\tau_R}}-\cancel{\frac{N'(t)}{\tau_R}}=\underbrace{\sigma_{21}\upsilon N_0}_{=\gamma}\Delta n'$$

$$\boxed{\frac{d\Delta n'}{dt}=-\underbrace{\left(\sigma_{21}\upsilon N_0+A_{21}+W_{03}\right)}_{=\alpha}\Delta n'-\underbrace{\frac{1}{\tau_R}}_{=\beta}N'}$$

$$\Rightarrow \begin{cases} \dfrac{d^2\Delta n'}{dt^2}+\alpha\dfrac{d\Delta n'}{dt}+\beta\gamma\Delta n'=0 \\ \dfrac{d^2 N'}{dt^2}+\alpha\dfrac{dN'}{dt}+\beta\gamma N'=0 \end{cases} \Rightarrow \begin{cases} \Delta n'(t)=\Delta n'(0)e^{-\varphi t}\sin\Omega_R t \\ N'(t)=N'(0)e^{-\varphi t}\sin\left(\Omega_R t-\dfrac{\pi}{2}\right) \end{cases}$$

$t=0 \Leftrightarrow \Delta n \to \Delta n_t$

- $t>0$ $\Delta n'(t)$, $\Delta N'(t)$ show damped oscillation with the same damping period

Attenuation coefficient

$$\varphi=\frac{\alpha}{2}=\frac{1}{2}\left(W_{03}+A_{21}+\sigma_{21}\upsilon N_0\right)$$

Damped oscillation frequency

$$\Omega_R=\frac{1}{2}\sqrt{4\beta\gamma-\alpha^2}=\sqrt{\beta\gamma-\varphi^2}$$

$$\xrightarrow[A_{21}\Delta n_t \approx (W_{03})_t\, n]{n >> \Delta n_t,\, 1/\tau_R >> W_{03}}$$

$$\boxed{\begin{aligned} \varphi &= \frac{1}{2}\sigma_{21}\upsilon\tau_R W_{03} n \\ \Omega_R &\approx \sqrt{\frac{A_{21}}{\tau_R}\left[\frac{W_{03}}{(W_{03})_t}-1\right]} \end{aligned}}$$

- $W_{03}\uparrow \rightarrow \varphi\uparrow,\ \Omega_R\uparrow$
- Multi-mode: For each mode, amplitude, φ and Ω_R are different. Therefore, random superposition and random fluctuation of intensity exist.

$$\Delta n'(t) = \Delta n'(0) e^{-\varphi t} \sin \Omega_R t$$

$$N'(t) = N'(0) e^{-\varphi t} \sin\left(\Omega_R t - \frac{\pi}{2}\right)$$

$t >> 1/\varphi \quad \Delta n'(t), N'(t) \to 0 \quad N'(t) \to N_0, \quad \Delta n'(t) \to (\Delta n)_0 = \Delta n_t$ ⇨ steady state

❖ Practical significance of relaxation oscillation (pros and cons)

- The relaxation oscillation reflects the instability of the laser oscillation process. It is universal. The relaxation oscillation is the effect of laser oscillation on the transition from a non-steady-state to a steady-state.

- Relaxation oscillation in CW lasers will lead to instability of laser output, which is called noise.

✓ warm-up time

✓ suddenly change the pump and loss will lead to oscillation instability

4.5 Line-width Limit of Single Mode Lasers

monochromatic ~ temporal coherence

❖ Passive cavity line-width: line-width of the eigen cavity mode ~ photon lifetime

Passive cavity (*g*=0: equal number of photons absorbed and emitted)
Intensity variation of light propagating in a passive cavity

$$I(t) = I_0 e^{-t/\tau_R} \qquad \tau_R = \frac{L'}{c\delta}$$

$$\Rightarrow \boxed{\Delta\nu_c = \frac{1}{2\pi\tau_R} = \frac{c\delta}{2\pi L'}}$$

Example: He-Ne laser, L=30 cm, r_1=1, r_2=0.98, $a \approx 0$, η=1.

$$\tau_R = 10^{-7} s, \quad \Delta\nu_c = 1.6\times10^{6} s^{-1}$$

❖ Active cavity linewidth (g>0)

$$\delta_s = \delta - g(\nu, I_\nu)l \Rightarrow \Delta\nu_s = \frac{c\delta_s}{2\pi L'} \Rightarrow \boxed{\Delta\nu_s < \Delta\nu_c} \qquad \tau'_R = L'/(c\delta_s)$$

Starting from the photon lifetime $\Delta\nu_s \to 0$?

linewidth of the active cavity is smaller than passive one

➢ Spontaneous emission always exits in the process of oscillation, laser linewidth can NOT be 0

Stimulated Emission	+	Spontaneous Emission	=	Loss
Coherent		Noncoherent		

$$P_0 = P_{sp} + P_{ste}$$

$$gl < \delta \qquad \delta_s \neq 0$$

✓ Existence of spontaneous emission → loss is not zero in active cavity

✓ Noncoherent spontaneous emission → random phase → a finite and slightly attenuated wave train of coherent radiation

✓ The line-width of the active cavity is determined by the net loss

❖ Estimation of $\Delta\nu_s$ (Four-level system, single mode, $L=l$) $\Delta\nu_s = \dfrac{c\delta_s}{2\pi L'}$

1. Get δ_s

$$\frac{dN_l}{dt} = \underbrace{\Delta n\sigma_{21}(\nu,\nu_0)}_{=\,g(\nu,I_\nu)}\upsilon N_l + \boxed{a_l n_2} - \frac{N_l}{\tau_R}$$

$$a_l = \frac{A_{21}\tilde{g}(\nu,\nu_0)}{n_\nu SL} = \frac{\sigma_{21}(\nu,\nu_0)\upsilon}{SL}$$ SP probability distributed to the mode l

stable oscillation

$$g(\nu,I_\nu)\upsilon N_l + a_l n_2 - \frac{N_l\delta\upsilon}{L} = 0 \qquad n_2 = n_{2t} \quad \Delta n = \Delta n_t$$

$$\delta_s = \delta - g(\nu,I_\nu)l = a_l n_{2t}\frac{L}{\upsilon N_l}$$

net loss $\neq 0$

compensated by SP

2. Get a_l, N_l

$$P_{out} = P_{sp} + P_{ste} \approx P_{ste}$$
$$= \Delta n_t\sigma_{21}(\nu,\nu_0)\upsilon N_l SLh\nu$$
$$= \Delta n_t a_l N_l (SL)^2 h\nu$$

$$P_{out} = \frac{N_l}{2}\upsilon Sh\nu T$$

$$\Downarrow$$

$$\Rightarrow \quad N_l = \frac{2P_{out}}{T\upsilon Sh\nu} \quad\Rightarrow\quad a_l = \frac{T\upsilon}{2SL^2\Delta n_t}$$

3. Theoretical line-width limit of a single mode laser

insert N_l, a_l into δ_s $\Longrightarrow$ insert δ_s

$$\boxed{\Delta \nu_s = \frac{n_{2t}}{\Delta n_t} \frac{2\pi (\Delta \nu_c)^2 h\nu}{P_{out}} \approx \frac{n_{2t}}{\Delta n_t} \frac{2\pi (\Delta \nu_c)^2 h\nu_0}{P_{out}}}$$

Example: He-Ne, λ=632.8 nm, L=30 cm, T=0.02, P_{out}=1 mW

$\Delta \nu_c = 1.6 \times 10^6$ Hz, $\Delta \nu_s = 6 \times 10^{-3}$ Hz, $n_{2t}/\Delta n_t = 1$

❖ The line-width of the real lasers

➢ Many orders of magnitude far beyond theoretical line-width limit

Reason: Frequency drift caused by temperature and other factors

- ✓ He-Ne laser (glass shell) 0.01 ^{o}C $\rightarrow$ 1 MHz
- ✓ Solid-state laser $\Delta T \rightarrow \nu_0$ shift

Summary $\Delta\nu_c = \dfrac{1}{2\pi\tau_R} = \dfrac{c\delta}{2\pi L'}$ $\Delta\nu_s = \dfrac{c\delta_s}{2\pi L'} = \dfrac{c a_l n_{2t} L/\upsilon N_l}{2\pi L'} = \dfrac{n_{2t}}{\Delta n_t}\dfrac{2\pi(\Delta\nu_c)^2 h\nu_0}{P_{out}}$

- Physical mechanism of line-width (single mode laser): SP, noise
- $\Delta\nu_s << \Delta\nu_c$
- $P_{out}\uparrow \rightarrow \Delta\nu_s\downarrow$ intra-cavity coherent photon number $\uparrow$, STE accounts for a larger proportion of P_{out}

$$\bar{n} = \frac{\rho_\nu}{8\pi h\nu^3/c^3} = \frac{B_{21}}{A_{21}}\rho_\nu = \frac{W_{21}}{A_{21}} \qquad \text{for He-Ne:} \quad \bar{n} \approx 3\times10^8$$

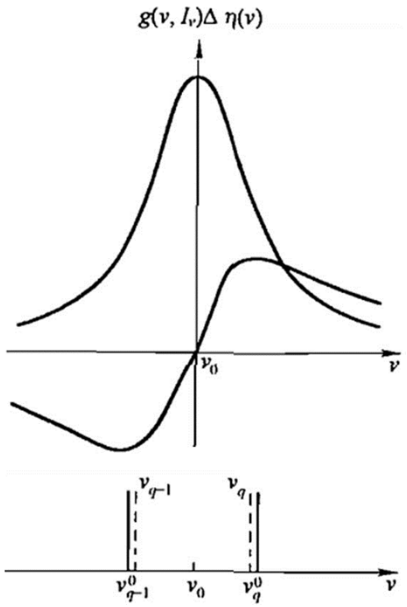
$$\bar{n} = \frac{\text{Total photon number}}{\text{Mode number}} \Rightarrow \frac{W_{21}}{\text{Total photon number}} = \frac{A_{21}}{\text{Mode number}}$$

- $\Delta\nu_s \propto (\Delta\nu_c)^2$
  - ✓ $\delta\downarrow \rightarrow \Delta\nu_c\downarrow \rightarrow \Delta\nu_s\downarrow$
  - ✓ $L\uparrow \rightarrow \Delta\nu_c\downarrow \rightarrow \Delta\nu_s\downarrow$ vertical external cavity semiconductor laser
- The actual line-width of a single-mode laser highly depends on the frequency stability

4.6 Frequency Pulling of Laser

论述

- What is frequency pulling?
- Physical mechanism: dispersion $$\eta = \eta_0 \left[1 - \frac{(\nu_0 - \nu) c}{\Delta \nu_H \omega} g \right]$$
- Dispersion induced frequency pulling $$\Delta\eta(\nu) = \frac{c(\nu - \nu_0)}{2\pi\nu\Delta\nu_H} g_H(\nu, I_\nu)$$



Passive cavity longitudinal frequency

$$\nu_q^0 = q\frac{c}{2\eta^0 L}$$

Active cavity longitudinal frequency

$$\nu_q = q\frac{c}{2\eta L} = q\frac{c}{2(\eta^0 + \Delta\eta)L}$$

Frequency difference between active and passive cavity

$$\nu_q - \nu_q^0 = q\frac{c}{2(\eta^0 + \Delta\eta)L} - q\frac{c}{2\eta^0 L}$$

$$\approx \frac{-\Delta\eta}{\eta^0}\nu_q^0$$

In real lasers, the dispersion of gain material makes longitudinal frequency approach more closer to the central frequency compared to the passive cavity: **Frequency Pulling**

➢ Discuss the refractive index change and pulling parameter (homogeneous and inhomogeneous broadening)

homogeneous broadening

$$\Delta\eta(\nu) = \frac{c(\nu-\nu_0)}{2\pi\nu\Delta\nu_H} g_H(\nu, I_\nu)$$

$$g_H(\nu, I_\nu) = \frac{A_{21}\upsilon\Delta n^0}{4\pi^2\nu_0^2\Delta\nu_H} \frac{(\Delta\nu_H/2)^2}{(\nu-\nu_0)^2 + (\Delta\nu_H/2)^2(1+I_\nu/I_s)}$$

frequency difference between active and passive cavity

$$\nu_q - \nu_q^0 \approx \frac{-\Delta\eta(\nu_q)}{\eta^0}\nu_q^0 \xRightarrow{\nu_q \approx \nu_q^0} \nu_q - \nu_q^0 \approx \frac{-c(\nu_q-\nu_0)}{2\pi\eta^0\Delta\nu_H} g_H(\nu_q, I_{\nu_q})$$

$\Downarrow$ steady-state

$$\nu_q - \nu_q^0 = \frac{-c(\nu_q-\nu_0)}{2\pi\eta^0\Delta\nu_H}\frac{\delta}{L}$$

$$\nu_q - \nu_q^0 = -\frac{\Delta\nu_c}{\Delta\nu_H}(\nu_q-\nu_0) \xLeftarrow{\Delta\nu_c = \frac{c\delta}{2\pi\eta^0 L}}$$

$\Downarrow$

$$\boxed{\sigma_H = -\frac{\nu_q - \nu_q^0}{\nu_q - \nu_0} = \frac{\Delta\nu_c}{\Delta\nu_H}}$$

pulling parameter of homogeneously broadened laser

inhomogeneous broadening

$$\Delta\eta_i(\nu) = \frac{c}{2\pi^{3/2}\nu} g_i^0(\nu_0) e^{-4\ln 2\left(\frac{\nu-\nu_0}{\Delta\nu_D}\right)^2} \int_0^{2\sqrt{\ln 2}\frac{\nu-\nu_0}{\Delta\nu_D}} e^{t^2} dt$$

$$\Delta\eta_i(\nu) = \frac{c(\nu-\nu_0)}{\pi^{3/2}\nu} \sqrt{\ln 2}\, g_i^0(\nu_0) e^{-4\ln 2\left(\frac{\nu-\nu_0}{\Delta\nu_D}\right)^2}$$

$$\nu_q - \nu_q^0 = -\frac{c(\nu_q-\nu_0)}{\pi^{3/2}\eta^0\Delta\nu_D} \sqrt{\ln 2}\, g_i^0(\nu_0) e^{-4\ln 2\left(\frac{\nu-\nu_0}{\Delta\nu_D}\right)^2}$$

$$= -\frac{c(\nu_q-\nu_0)}{\pi^{3/2}\eta^0\Delta\nu_D} \sqrt{\ln 2}\, g_i(\nu_q, I_{\nu_q}) \sqrt{1+\frac{I_{\nu_q}}{I_s}}$$

Steady state $g_i(\nu_q, I_{\nu_q}) = \frac{\delta}{L}$

$$= -2\sqrt{\frac{\ln 2}{\pi}} \frac{\Delta\nu_c}{\Delta\nu_D} \sqrt{1+\frac{I_{\nu_q}}{I_s}} (\nu_q - \nu_0)$$

$$\because \Delta\nu_c = \frac{c\delta}{2\pi\eta^0 L}$$

$$\boxed{\sigma_i = -\frac{\nu_q - \nu_q^0}{\nu_q - \nu_0} = 2\sqrt{\frac{\ln 2}{\pi}} \frac{\Delta\nu_c}{\Delta\nu_D} \sqrt{1+\frac{I_{\nu_q}}{I_s}}}$$

He-Ne laser

$\sigma_i = 10^{-3}$